

LABORATORY REPORT SHEET

EXPERIMENT: A0 PRECISION MEASUREMENTS

REG. No. F17/14290/2021
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OBSERVATIONS AND DATA ANALYSIS

Results

Worksheet 1

NB. You must include in the table the units of measurements for any measurements you take



Object	Measurements				
	Dimensions	Metre Rule	Vernier Calipers	Micrometer screw gauge	Weighting Balance
	Zero Error	± 0	± 0.000	± 0.000	± 0.0
	Reading Error	$\pm 0.1 \text{ cm}$	$\pm 0.01 \text{ mm}$	$\pm 0.01 \text{ mm}$	$\pm 0.1 \text{ g}$
Aluminum Cylinder	Height	15.0 cm			50.3 g
	Diameter	1.2 cm	1.23 cm		
Steel Ball	Diameter			19.49 mm	3.4 g
Copper Wire	Length	7.0 cm	7.01 cm		0.6 g
	Diameter	0.1 cm		1.14 mm	
Glass Capillary Tube	Length	7.2 cm	7.21 cm		
	Diameter	0.6 cm	0.63 cm	6.64 mm	13.3 g

Internal diameter of capillary tube using traveling microscope = $0.20 \text{ cm} \pm \dots$

Table 2

Object	Volume (m^3)	Error	Density	Error
Copper Cylinder	$1.697142857 \times 10^{-5}$	$5.77085714 \times 10^{-4}$	$2963.804714 \text{ kg/m}^3$	$5.892265971 \text{ kg/m}^3$
Copper Wire	7.219278×10^{-8}	$2.62591429 \times 10^{-9}$	$2311.020416 \text{ kg/m}^3$	$1385.12009 \text{ kg/m}^3$
Steel Ball	$4.47844685 \times 10^{-7}$	$2.83046026 \times 10^{-9}$	$7594.634702 \text{ kg/m}^3$	$222.371629 \text{ kg/m}^3$
Glass Capillary Tube	1.5890688×10^{-4}	7.132×10^{-1}	83.69651 kg/m^3	45230973 kg/m^3

Table 3

Instrument	Current	Error	Unit	
Ammeter	0.52	± 0.01	A	
Milliammeter	1.8	± 0.2	mA	
Microammeter	19	± 1	μA	

Analysis

Volume of Copper Cylinder

$$\text{Volume} = \pi r^2 h$$

$$= \left(\frac{22}{7} \times 0.006^2 \times 0.15 \right) m^3$$

$$= 1.697142857 \times 10^{-5} m^3$$

Density

$$\text{Density} = \frac{\text{Mass}}{\text{Volume}}$$

$$= \frac{0.0503 \text{ kg}}{1.697142857 \times 10^{-5} m^3}$$

$$= 2963.804714 \text{ kg/m}^3$$

Errors in volume

$$\left(\frac{0.1}{0.006} + \frac{0.1}{0.006} + \frac{0.1}{0.15} \right) \times 1.697142857 \times 10^{-5}$$

$$= 5.77085714 \times 10^{-4} m^3$$

Error in density

$$\left(\frac{0.1}{50.3} \right) + \left(\frac{5.77085714 \times 10^{-4}}{1.697142857 \times 10^{-5}} \right) \times 2963.804714 \text{ kg/m}^3$$

$$= 5.892265971 \text{ kg/m}^3$$

Volume of Copper wire

$$\begin{aligned} \text{vol} &= \pi r^2 h \\ &= \frac{22}{7} \times \left(\frac{0.57}{1000}\right)^2 \times \frac{7.07}{100} \\ &= 7.219278 \times 10^{-8} \text{ m}^3 \end{aligned}$$

Density of copper wire

$$\begin{aligned} D &= \frac{\text{Mass}}{\text{volume}} \\ &= \left(\frac{0.69}{1000}\right) \text{ kg} \\ &\quad \frac{1}{7.219278 \times 10^{-8} \text{ m}^3} \\ &= 8311.080416 \text{ kg/m}^3 \end{aligned}$$

Error in Volume

$$\begin{aligned} &\left(\left(\frac{0.01}{0.57}\right) + \left(\frac{0.01}{0.57}\right) + \left(\frac{0.01}{7.07}\right) \right) \\ &\quad \times 7.219278 \times 10^{-8} \text{ m}^3 \\ &= 2.635191429 \times 10^{-9} \text{ m}^3 \end{aligned}$$

Error in density

$$\begin{aligned} &= \left(\frac{0.1}{0.6}\right) + \left(\frac{2.635191429 \times 10^{-9}}{7.219278 \times 10^{-8}}\right) \\ &\quad \times 8311.080416 \text{ kg/m}^3 \\ &= 1385.120069 \text{ kg/m}^3 \end{aligned}$$

Volume of steel ball

$$\begin{aligned} \text{Vol} &= \frac{4}{3} \pi r^3 \\ &= \frac{4}{3} \times \frac{22}{7} \times \left(\frac{4.745}{1000}\right)^3 \\ &= 4.476844685 \times 10^{-7} \text{ m}^3 \end{aligned}$$

Density of steel ball

$$\begin{aligned} \text{Density} &= \frac{\text{Mass}}{\text{volume}} \\ &= 3.4 \times 10^{-3} \\ &\quad \frac{4.476844685 \times 10^{-7}}{7594.634702 \text{ kg/m}^3} \end{aligned}$$

Error in volume

$$\begin{aligned} &\left(\left(\frac{0.01}{4.745}\right) + \left(\frac{0.01}{4.745}\right) + \left(\frac{0.01}{4.745}\right) \right) \\ &\quad \times 4.476844685 \times 10^{-7} \\ &= 2.830460286 \times 10^{-9} \end{aligned}$$

Error in density

$$\begin{aligned} &\left(\left(\frac{0.1}{3.4}\right) + \left(\frac{2.830460286 \times 10^{-9}}{4.476844685 \times 10^{-7}}\right) \right) \\ &\quad \times 7594.634702 \text{ kg/m}^3 \\ &= 223.3716089 \text{ kg/m}^3 \end{aligned}$$

Volume of Glass Capillary Tube

$$\begin{aligned} V &= \pi h (R^2 - r^2) \\ &= \frac{22}{7} \times 1.2 \left((3.32 \times 10^{-3})^2 - (2 \times 10^{-3})^2 \right) \\ &= 1.5890688 \times 10^{-4} \text{ m}^3 \end{aligned}$$

Density of glass capillary Tube

$$\begin{aligned} \text{Density} &= \frac{\text{Mass}}{\text{Volume}} \\ &= \frac{13.3}{1000} \text{ kg} \\ &\quad \frac{1}{1.5890688 \times 10^{-4} \text{ m}^3} \\ &= 83.69681665 \text{ kg/m}^3 \end{aligned}$$

Error in Volume

$$\begin{aligned} dV &= \left[\left(\frac{0.01}{7.21} \right) \cdot 2 + \left(\frac{0.1}{0.212} \right) \right] \\ &\times 6029.072^{-6} \\ &= 7.132 \times 10^{-7} \text{ m}^3 \end{aligned}$$

Error in Density

$$\begin{aligned} dp &= \left[\left(\frac{0.1}{13.3} \right) + \left(\frac{7.132 \times 10^{-7}}{2.206 \times 10^{-6}} \right) \right] \\ &\times 6029.012 \\ &= 45.3309173 \text{ kg/m}^3 \end{aligned}$$

Questions



1. Why is it appropriate to use the meter rule for measuring the length of the copper wire but the micrometer screw gauge for the diameter?

A meter rule can measure distances of upto a meter more precisely. Micrometer screw gauge measures the diameter more accurately than the other instruments. Its smallest unit of measurement is 0.01 mm .

2. What is the difference between accuracy and precision?

Accuracy is the proximity of the measurement result to the true value. Precision is the ability of an instrument to give the same value everytime measurement is repeated.

Discussion

Conclusions

The volume of the copper cylinder was found to be $1.697142857 \times 10^{-5} \text{ m}^3$ and its density was found to be $2963.804714 \text{ kg/m}^3$. Write similar conclusions for the steel ball and capillary tube and compare your values with the correct values for the densities of copper and steel and comment.

The volume of the steel ball was found to be $4.476844685 \times 10^{-7} \text{ m}^3$ and its density was found to be $7594.634702 \text{ kg/m}^3$. The volume of capillary tube was $1.5890668 \times 10^{-6} \text{ m}^3$ and its density was 83.69681 kg/m^3 . The density of aluminium is 2710 kg/m^3 while that of steel is 8050 kg/m^3 . The difference in densities is due to errors during measurements both from the instrument and errors due to physical conditions such as temperature.

References